EXPLORATION OF PHYSIOLOGICAL RHYTHMS USING GENETIC ALGORITHMS

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Introduction

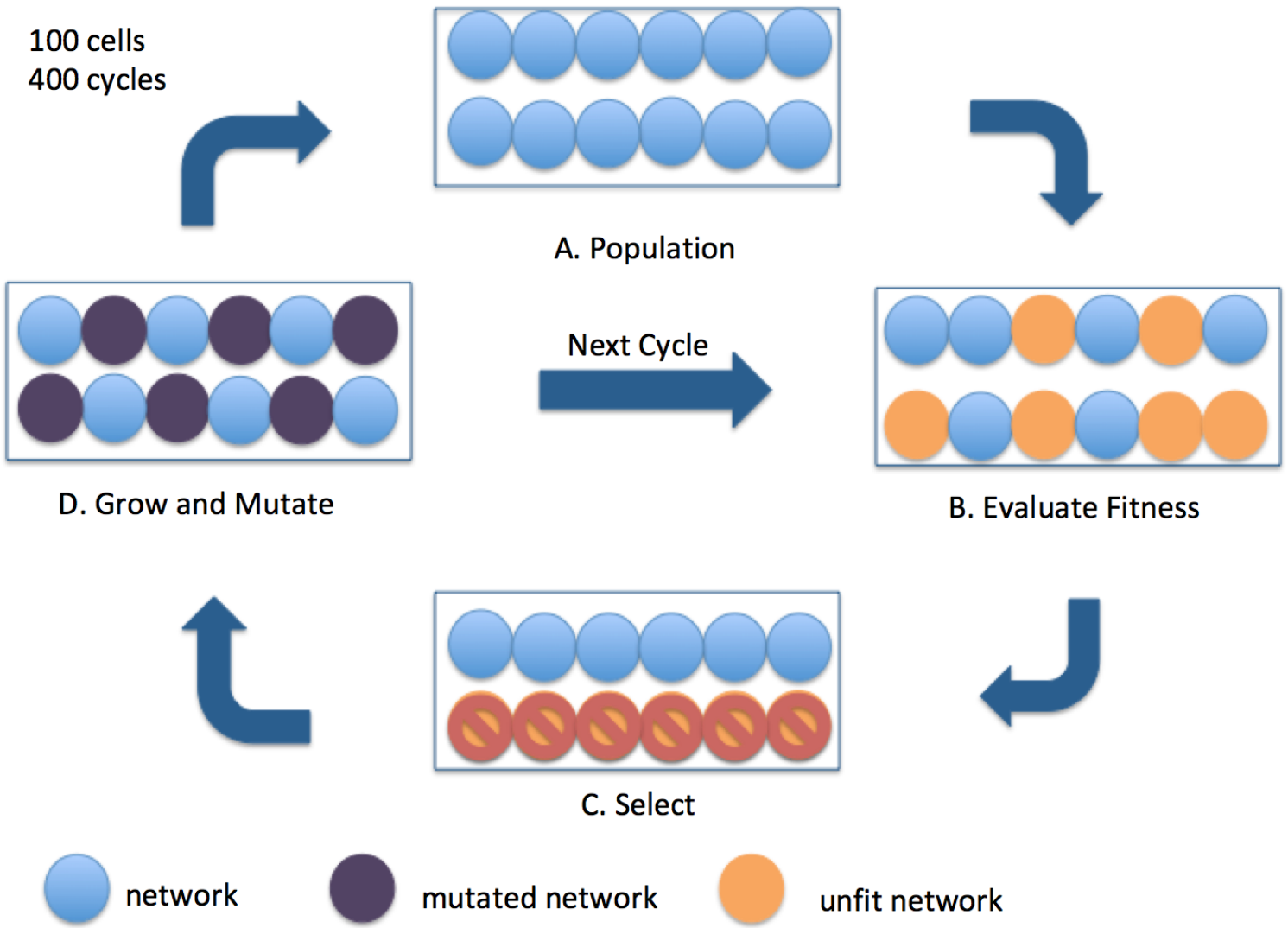
- All organisms have internal clocks, like circadian rhythms, reproductive rhythms, etc.
- Those biological clocks are related to the pattern of some specific molecules.
- We now understand mechanisms in some simple organism, but the genetic and molecular of most biological clocks are still mysterious.

Project Goals

- Our project uses genetic algorithms to model the molecular clocks through a collection of ordinary differential equations.[1]
- We will interpret those ODEs biologically and compare with the known results to confirm our understanding.

[1] *Source: P. Francois, V.Hakim. Design of genetic networks with specified functions by evolution in silico (2003)*

General Algorithm - an evolutionary procedure *in silico*



General Algorithm - work done so far

- Investigated possible mutations described by altered interactions (ex. gene-protein or protein-protein) in one cell
- Implemented initialization and mutation code in Python
- Approximated gene-protein interaction by using Michaelis-Menten Kinetics

Future Direction

- Finish implementing Python codes
  - Solve ODEs in Python
  - Apply Fourier Transform to construct a fitness function
- Classify different ODEs and compare them to real biological examples